

Week 8: Transformers and Impedance Matching

Ideal transformer, turns ratio, dot convention, reflected impedance, matching, real transformers

In-depth notes covering all core material of this chapter: theory, engineering shortcuts, exam traps, 7 worked examples, and 8 practice problems with full solutions.

Based on: Rizzoni & Kearns, *Principles and Applications of Electrical Engineering*, 7th ed., §13.4 and §14.4

Inspired by: ELEC 275, Principles of Electrical Engineering, Concordia University

By **Vladislav Shkuratov**, made with Claude (AI). Independent study notes, not affiliated with Concordia University or any instructor.

Contents

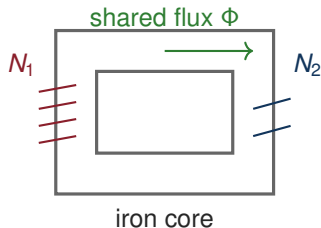
PART 1: Lecture notes	4–33
What a transformer is, symbols	4–5
The ideal transformer: voltage, current, power	6–10
Dot convention and polarity	11–14
Reflected impedance	15–18
Impedance matching for maximum power	19–22
Circuits with transformers, exam-style problem	23–27
Real transformers and efficiency	28–30
Traps and exam tips	31–32
Summary	33–33
PART 2: Final check (8 practice problems)	34–42
PART 3: Answer sheet (full solutions)	43–51

How to use this deck

- **“Try it” slides:** the problem alone, with empty space. Do it first.
- **Next slide:** the clean solution, step by step.
- **Purple boxes:** engineering shortcuts, the fast way to see or solve things.
- **Orange boxes:** traps. This is where marks get lost on exams.
- Formulas are tagged **[on formula sheet]** or **[NOT on formula sheet]**, based on the sheet given in ELEC 275 exams.
- **End of deck:** final-check problems, then a full answer sheet.

The right side of each slide is left empty for your notes.

What a transformer is



Two coils wound on one iron core. AC current in the **primary** (N_1 turns) makes a changing flux. The core carries that flux through the **secondary** (N_2 turns), which induces a voltage there.

No wire connects the two sides. Energy crosses through the magnetic field.

Uses

Change voltage levels (power grid, chargers), isolate circuits for safety, and match impedances (audio amplifiers to speakers).

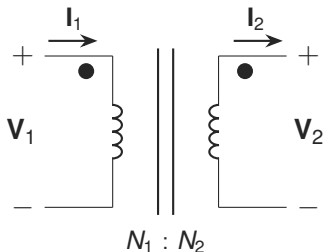
Symbols for this week

Symbol	Meaning
N_1, N_2	number of turns on the primary (source side) and secondary (load side)
$N_1 : N_2$	turns ratio, written primary first
$n = N_2/N_1$	the ratio used in this deck: $n > 1$ steps voltage up , $n < 1$ steps it down
V_1, I_1	primary voltage and current (current flows into the primary dot)
V_2, I_2	secondary voltage and current (current flows out of the secondary dot)
Z'_L	the load impedance as seen from the primary (“reflected”)

Trap: books and exams define “ n ” or “ N ” differently (N_2/N_1 or N_1/N_2). Check which one a problem means. Writing $N_1 : N_2$ avoids the confusion.

The ideal transformer

[NOT on formula sheet: memorize]



Voltage $\frac{V_2}{V_1} = \frac{N_2}{N_1} = n$

Current $\frac{I_2}{I_1} = \frac{N_1}{N_2} = \frac{1}{n}$

Power $V_1 I_1^* = V_2 I_2^* \text{ (in = out)}$

Engineering shortcut: voltage follows turns, current goes the other way

More turns means more voltage and **less** current. Voltage up $\times 10$ means current down $\times 10$: same power.

Where “ideal” comes from

An ideal transformer assumes: no winding resistance, no leakage flux, infinite core permeability (no magnetizing current), and no core loss.

Why the voltage ratio is the turns ratio

Both coils see the same flux Φ . Faraday: $v_1 = N_1 \frac{d\Phi}{dt}$, $v_2 = N_2 \frac{d\Phi}{dt}$, so

$$\frac{v_2}{v_1} = \frac{N_2}{N_1}.$$

Trap

Transformers only work with **AC**. DC makes a constant flux, $d\Phi/dt = 0$, so nothing is induced.

Why the current goes the other way

Ampere-turn balance: in an ideal core ($\mu \rightarrow \infty$) it takes zero MMF to hold the flux. So the two windings' ampere-turns must cancel:

$$N_1 I_1 = N_2 I_2 \quad \Rightarrow \quad \frac{I_2}{I_1} = \frac{N_1}{N_2} = \frac{1}{n}$$

Same answer from energy: an ideal transformer loses nothing, so power in = power out: $V_1 I_1 = V_2 I_2$. Since $V_2 = nV_1$, the current must be $I_2 = I_1/n$.

Think of it as a lever: more turns means more voltage and proportionally less current, exactly like a longer lever arm gives more distance and proportionally less force.

Try it: Example 1, basic ratios

A 10:1 transformer ($N_1 : N_2$) has 120 V RMS on the primary. A $4\ \Omega$ resistor is on the secondary.

Find V_2 , I_2 , I_1 , and the power on each side.

your work here ↓

Example 1: solution

Ratio 10:1 means $N_1 = 10$, $N_2 = 1$ (in proportion):
 $n = N_2/N_1 = 1/10$, a step-down

V_2 voltage follows the turns: $V_2 = nV_1 = 120/10 = 12 \text{ V}$

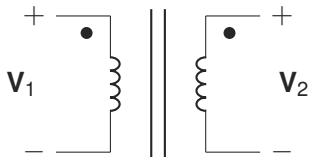
I_2 Ohm's law on the secondary: $I_2 = V_2/R = 12/4 = 3 \text{ A}$

I_1 current goes by the inverse ratio: $I_1 = nI_2 = 3/10 = 0.3 \text{ A}$

Check power in = $120 \times 0.3 = 36 \text{ W}$ = power out = $12 \times 3 = 36 \text{ W}$
✓

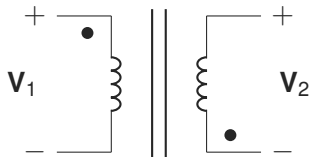
The dot convention

[NOT on formula sheet: memorize]



dots at the same end:

$$V_2 = +nV_1$$



dots at opposite ends:

$$V_2 = -nV_1$$

The rule

The dotted terminals are + at the same moment. If V_1 is defined + at its dot and V_2 is defined + at its dot, then $V_2 = nV_1$. If one voltage is defined against its dot, add a minus sign (180° shift).

What the dots actually mean

A dot marks the end of each winding where current **entering** would push the flux in the **same direction** around the core.

The two coils could be wound either way round the core (clockwise or counterclockwise). The dots record which way, without drawing the core.

Consequence: a changing flux induces voltages that are $+$ at the dotted ends at the same instant. So if both voltages are defined $+$ at their dots, they're in phase: $V_2 = nV_1$. If one is defined the other way, it's upside down: $V_2 = -nV_1$ (180°).

Current version: current **into** the primary dot comes **out of** the secondary dot. That's the direction convention used for I_2 in this deck.

Try it: Example 2, dot convention

Same transformer as Example 1 (10:1, 120 V RMS $\angle 0^\circ$ on the primary, + at the primary dot). The secondary dot is at the **bottom**, but $\mathbf{V}_2$ is measured with + at the **top**.

Find $\mathbf{V}_2$ as a phasor.

your work here ↓

Example 2: solution

Size the magnitude only depends on the ratio: $|\mathbf{V}_2| = 120/10 = 12$
V

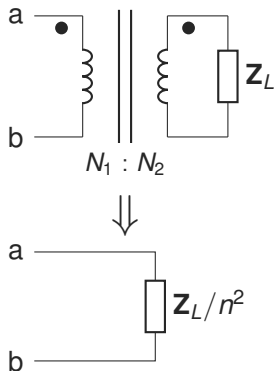
Sign primary voltage is defined + at its dot; the secondary's + is at the top, but its dot is at the bottom, so it's measured **against** its dot: multiply by -1

Result $\mathbf{V}_2 = -\frac{1}{10}(120\angle 0^\circ) = -12 = 12\angle 180^\circ \text{ V}$

Shortcut: dots never change magnitudes, only the sign (phase). If a question asks only for sizes or power, dots don't matter.

Reflected impedance

[NOT on formula sheet: memorize]



$$\mathbf{Z}'_L = \frac{\mathbf{Z}_L}{n^2} = \left(\frac{N_1}{N_2} \right)^2 \mathbf{Z}_L$$

Seen from the primary, the load looks like $\mathbf{Z}_L$ scaled by the **square** of the turns ratio. Then remove the transformer and solve a normal circuit.

Shortcut: secondary $\rightarrow$ primary: impedances $\times (N_1/N_2)^2$. Primary $\rightarrow$ secondary: impedances $\times (N_2/N_1)^2$ and voltages $\times N_2/N_1$.

Where $\mathbf{Z}_L/n^2$ comes from

The primary sees $\mathbf{Z}' = \mathbf{V}_1/\mathbf{I}_1$. Write both in secondary quantities using the two transformer rules:

$$\mathbf{V}_1 = \frac{\mathbf{V}_2}{n}, \quad \mathbf{I}_1 = n \mathbf{I}_2$$

$$\mathbf{Z}' = \frac{\mathbf{V}_1}{\mathbf{I}_1} = \frac{\mathbf{V}_2/n}{n \mathbf{I}_2} = \frac{1}{n^2} \frac{\mathbf{V}_2}{\mathbf{I}_2} = \frac{\mathbf{Z}_L}{n^2} = \left(\frac{N_1}{N_2} \right)^2 \mathbf{Z}_L$$

The ratio appears **twice** (once from the voltage, once from the current), which is why impedance scales with its square.

Step-down (more turns on the primary) makes a small load look **bigger** to the source. An $8 \, \Omega$ speaker behind a 5:1 transformer looks like $200 \, \Omega$.

Try it: Example 3, reflect a speaker

An $8\ \Omega$ speaker is on the secondary of a 5:1 ($N_1 : N_2$) transformer.

What impedance does the amplifier on the primary see?

your work here ↓

Example 3: solution

Rule load moved to the primary side: multiply by the **square** of N_1/N_2

Ratio 5:1, so $N_1/N_2 = 5$

Result $Z' = 5^2 \times 8 = 25 \times 8 = 200 \, \Omega$

Trap: $5 \times 8 = 40$ is the classic mistake: the ratio appears twice (voltage and current), so it's squared.

Impedance matching

[NOT on formula sheet: memorize]

Goal

A source with resistance R_s drives a load R_L that is the “wrong” size. Pick the turns ratio so the **reflected** load equals R_s (max power, week 3):

$$\left(\frac{N_1}{N_2}\right)^2 R_L = R_s \quad \Rightarrow \quad \frac{N_1}{N_2} = \sqrt{\frac{R_s}{R_L}}$$

With reactance (AC max power, week 7)

The transformer only scales. To cancel a reactance, add a series capacitor or inductor so the total reactance is zero, then match the resistances with the turns ratio.

Why matching works

From week 3, a source with internal resistance R_s gives the most power to a load that looks like R_s .

The transformer changes what the load **looks like** without wasting any power (ideal: power in = power out). So choose the ratio that makes the reflected load equal R_s :

$$\left(\frac{N_1}{N_2}\right)^2 R_L = R_s \quad \Rightarrow \quad \frac{N_1}{N_2} = \sqrt{\frac{R_s}{R_L}}$$

Whatever power reaches the reflected resistor on the primary side is exactly what the real load receives on the secondary side. **With reactance in the source:** the transformer scales R and X together, so it can't cancel reactance. Add a series C (or L) to cancel it, then match the resistances with the ratio (Example 6).

Try it: Example 4, matching an amplifier

An amplifier has $V_s = 20$ V RMS and internal resistance $R_s = 800 \Omega$. It drives an 8Ω speaker.

(a) Turns ratio for maximum power? (b) That power?

your work here ↓

Example 4: solution

Goal maximum power: the reflected speaker must look like
 $R_s = 800 \Omega$

(a) $(N_1/N_2)^2 \times 8 = 800 \Rightarrow N_1/N_2 = \sqrt{800/8} = \sqrt{100} =$
10 : 1

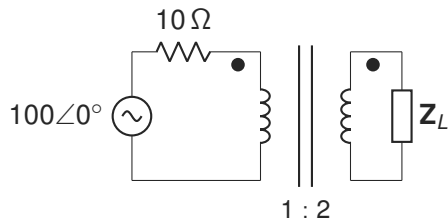
(b) the source now sees 800Ω , a matched load:

$$P_{max} = \frac{V_s^2}{4R_s} = \frac{20^2}{4 \times 800} = \mathbf{0.125 \text{ W}}$$

(all of it reaches the speaker)

Compare no transformer: $I = 20/(800 + 8) = 24.75 \text{ mA}$,
 $P = I^2 \times 8 = 4.9 \text{ mW}$: matching gives about 25 times more

Try it: Example 5, a circuit with a transformer



$N_1 : N_2 = 1 : 2$, dots on top,
source in V RMS,
 $Z_L = 40 + j40\ \Omega$.

Find $\mathbf{I}_1$, $\mathbf{V}_1$, $\mathbf{V}_2$, $\mathbf{I}_2$ and the
load power.

your work here ↓

Example 5: solution (1/2)

Plan reflect the load to the primary, solve that one-loop circuit, then step back to the secondary

Reflect 1:2 means $N_1/N_2 = 1/2$:
 $\mathbf{Z}'_L = (1/2)^2(40 + j40) = 10 + j10 \Omega$

I₁ loop: source, 10Ω , and $\mathbf{Z}'_L$:

$$\frac{100}{10 + 10 + j10} = \frac{100}{22.36 \angle 26.57^\circ} = 4.47 \angle -26.6^\circ \text{ A}$$

V₁ the primary's voltage is across $\mathbf{Z}'_L$:

$$\mathbf{I}_1 \mathbf{Z}'_L = 4.47 \angle -26.6^\circ \times 14.14 \angle 45^\circ = 63.2 \angle 18.4^\circ \text{ V}$$

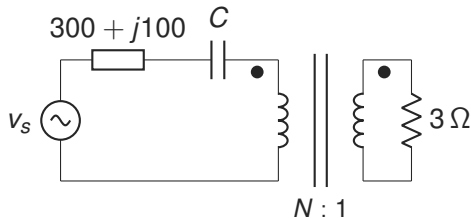
Example 5: solution (2/2)

V_2 step up by $n = 2$ (dots on top, same sign): $126.5 \angle 18.4^\circ \text{ V}$

I_2 current by the inverse: $I_1/2 = 2.24 \angle -26.6^\circ \text{ A}$

P_L only the load's 40Ω absorbs: $2.24^2 \times 40 = 200 \text{ W}$ (same as $4.47^2 \times 10$ on the primary ✓)

Try it: Example 6, exam style



$v_s = 120 \cos(377t)$ V,
source impedance
 $300 + j100 \Omega$.

Find N and C for maximum
power to the 3Ω load, and
that power.

your work here ↓

Example 6: solution

Plan two conditions for maximum power: cancel the reactance with C , then match the resistance with N

C the source has $+j100$, so the capacitor must give $-j100$:

$$\frac{1}{\omega C} = 100 \Rightarrow C = \frac{1}{(377)(100)} = 26.5 \mu\text{F}$$

N reflected load must equal the source's 300Ω : $N^2 \times 3 = 300$
 $\Rightarrow N = 10$ (10 : 1)

RMS v_s is given as a peak: $V_{rms} = 120/\sqrt{2} = 84.85 \text{ V}$

Power $P_{max} = \frac{V_{rms}^2}{4R} = \frac{84.85^2}{1200} = 6 \text{ W}$ (using 120 V peak would wrongly give 12 W)

Real transformers and efficiency

A real transformer adds:

Winding resistance → copper loss $I^2 R$
(grows with load).

Leakage inductance → some flux
misses the other coil.

Magnetizing current → the core needs
current to hold flux.

Core loss (hysteresis, eddy currents) →
roughly constant.

$$\eta = \frac{P_{out}}{P_{out} + P_{losses}}$$

Big power transformers reach
98–99%.

Rating

Transformers are rated in **kVA** (apparent power), not kW, because the heating depends on current, whatever the power factor. $P_{out} = |\mathbf{S}| \times \text{pf}$.

Try it: Example 7, efficiency

A 10 kVA transformer runs at full load with pf 0.8. Core loss is 100 W and copper loss is 200 W.

Find the output power and the efficiency.

your work here ↓

Example 7: solution

Output rating is apparent power: $P_{out} = |\mathbf{S}| \times \text{pf} = 10\,000 \times 0.8 =$
8000 W

Losses core (fixed) + copper (load-dependent): $100 + 200 = 300 \text{ W}$

$$\eta = \frac{P_{out}}{P_{out} + \text{losses}} = \frac{8000}{8300} = \mathbf{96.4\%}$$

Traps

Check these before you box an answer

- Check which ratio “ n ” or “ N ” means in the problem.
- Voltage scales by the ratio, current by its inverse, impedance by its **square**.
- Dots only change the sign (phase), never the magnitude.
- Transformers don't pass DC.
- Peak vs RMS in power calculations.
- Power in = power out (ideal). Use it as a check every time.

Exam-format traps

Show the reflected circuit

Draw the equivalent circuit with the load reflected to the primary. It shows your method and earns marks even if the arithmetic slips.

Two unknowns, two conditions

“Find N and C for maximum power”: C cancels the reactance, and N matches the resistance. Solve them separately.

Summary

Ideal transformer	$V_2/V_1 = N_2/N_1 = n$, $I_2/I_1 = 1/n$, power in = out
Dots	both voltages + at the dots: $V_2 = nV_1$; one against: add —
Reflection	$Z'_L = Z_L/n^2 = (N_1/N_2)^2 Z_L$
Matching	$N_1/N_2 = \sqrt{R_s/R_L}$; cancel reactance with a series C or L
Real	copper, core, leakage, magnetizing; $\eta = P_{out}/(P_{out} + \text{losses})$; rated in kVA

Final check

8 problems. One per slide, with space. No looking back.

If one takes more than 10 min, go back to that section. The answer sheet comes after the last problem.

P1: step down

A transformer steps 240 V RMS down to 24 V RMS. The primary has 500 turns. A $6\ \Omega$ load is on the secondary.

Find N_2 , I_2 and I_1 .

your work here ↓

P2: step up

A 1:5 transformer ($N_1 : N_2$) has 100 V RMS on the primary and a 1 k Ω load.

Find V_2 , I_2 , I_1 and the power delivered.

your work here ↓

P3: reflected impedance

A $4\ \Omega$ load is on the secondary of a 3:1 ($N_1 : N_2$) transformer. What does the primary see?

your work here ↓

P4: matching

A source with $V_s = 10 \text{ V RMS}$ and $R_s = 1.6 \text{ k}\Omega$ drives a $4 \text{ }\Omega$ load.

Turns ratio for maximum power, and that power?

your work here ↓

P5: circuit with a transformer

Source $50\angle 0^\circ$ V RMS with series $5\ \Omega$, then a 1:2 transformer (dots on top), then a load $20 - j20\ \Omega$.

Find $\mathbf{I}_1$, $\mathbf{V}_2$ and the load power.

your work here ↓

P6: dots

A 1:4 transformer has $\mathbf{V}_1 = 30\angle 20^\circ \text{ V}$ (+ at its dot). The secondary dot is at the bottom and $\mathbf{V}_2$ is measured + at the top.

Find $\mathbf{V}_2$.

your work here ↓

P7: efficiency

A 5 kVA transformer delivers 4 kW. Core loss is 80 W and copper loss is 120 W. Efficiency? Load power factor?

your work here ↓

P8: exam style

$v_s = 120\sqrt{2} \cos(377t)$ V (120 V RMS), source impedance $200 + j50 \Omega$, a series capacitor C , then an $N : 1$ transformer driving an 8Ω load.

Find N , C and P_{max} .

your work here ↓

Answer sheet

Full worked solutions to P1–P8.

P1: solution

Ratio voltage follows turns: $N_2/N_1 = V_2/V_1 = 24/240 = 1/10$

$$N_2 \quad 500 \times \frac{1}{10} = \boxed{50 \text{ turns}}$$

$$I_2 \quad \text{Ohm on the secondary: } 24/6 = \boxed{4 \text{ A}}$$

$$I_1 \quad \text{inverse ratio: } 4 \times \frac{1}{10} = \boxed{0.4 \text{ A}} \quad (\text{check: } 240 \times 0.4 = 24 \times 4 = 96 \text{ W } \checkmark)$$

P2: solution

V_2 1:5 step-up: $V_2 = 5 \times 100 = 500 \text{ V}$

I_2 $500/1000 = 0.5 \text{ A}$

I_1 current by the inverse: $0.5 \times 5 = 2.5 \text{ A}$ (more current on the low-voltage side)

P $500 \times 0.5 = 250 \text{ W} = 100 \times 2.5 \checkmark$

P3: solution

Rule secondary to primary: multiply by $(N_1/N_2)^2$

Result $N_1/N_2 = 3$: $3^2 \times 4 = 36 \Omega$

P4: solution

Match reflected load = $R_s: (N_1/N_2)^2 \times 4 = 1600 \Rightarrow$

$$N_1/N_2 = \sqrt{400} = \boxed{20 : 1}$$

Power source sees a matched 1600Ω : $P = \frac{V_s^2}{4R_s} = \frac{10^2}{6400} =$

$$\boxed{15.6 \text{ mW}}$$

P5: solution

Reflect $N_1/N_2 = 1/2$: $(1/2)^2(20 - j20) = 5 - j5 \Omega$

I₁ loop $5 + \frac{5 - j5}{50} = 10 - j5 = 11.18 \angle -26.57^\circ$:
 $\frac{11.18 \angle -26.57^\circ}{11.18 \angle -26.57^\circ} = 4.47 \angle 26.6^\circ \text{ A}$

V₁ across $\mathbf{Z}'_L$: $4.47 \angle 26.6^\circ \times 7.07 \angle -45^\circ = 31.6 \angle -18.4^\circ \text{ V}$

V₂ step up by 2: $63.2 \angle -18.4^\circ \text{ V}$

P_L real part of the reflected load: $|\mathbf{I}_1|^2 \times 5 = 20 \times 5 = 100 \text{ W}$

P6: solution

Size 1:4 step-up: $4 \times 30 = 120 \text{ V}$

Sign V_2 is measured against its dot: multiply by -1 , i.e. add 180° to the angle

Result $20^\circ + 180^\circ = 200^\circ$, same as -160° : $V_2 = 120 \angle -160^\circ \text{ V}$

P7: solution

$$\eta \quad \frac{P_{out}}{P_{out} + \text{losses}} = \frac{4000}{4000 + 80 + 120} = \frac{4000}{4200} = 95.2\%$$

$$\text{pf} \quad \text{at full rating, } |\mathbf{S}| = 5000 \text{ VA: } \text{pf} = P/|\mathbf{S}| = 4000/5000 = 0.8$$

P8: solution

C cancel the source's $+j50$: $\frac{1}{\omega C} = 50 \Rightarrow C = \frac{1}{(377)(50)} =$

$$53.1 \mu\text{F}$$

N reflected load = 200Ω : $N^2 \times 8 = 200 \Rightarrow N^2 = 25 \Rightarrow$

$$N = 5$$

Power source voltage is already RMS (120 V): $P_{max} = \frac{120^2}{4 \times 200} =$

$$18 \text{ W}$$